Supplementary evidence for Life manuscript 4620846

Modular CNN–Transformer Specialists for Multi-Source 12-Lead ECG Rhythm and Pathology Classification

Table S1: Output availability, recovered supervision and evaluation evidence for the eleven out-of-profile or auxiliary endpoints.

Class	Global logit	Recovered direct supervision	Published specialist evaluation	Later experimental evidence	Interpretation limit
NSR	Yes	V3F: P8 direct term documented. V3AG: Direct subset documented.	Native P8 endpoint; not P6	Native P8 evaluation. Selected development and compatible LUDB principal-rhythm references.	Not a residual arrhythmia or proof of a disease-free ECG
PAC	Yes	V3F: P8 direct term documented. V3AG: Not in the recovered eight-target subset.	Native P8 endpoint; not P6	Native P8 and experimental P6 Other union. SPH reference mapping; grouped P6 residual head.	ECG statement recognition, not beat-by-beat detection
Flutter	Yes	V3F: No direct term in recovered stage. V3AG: Direct subset and internal panels documented.	No native P6/P8 endpoint	New SPH binary head evaluated. Held source-specific contrast; not released.	Internal V3AG panels are not published specialist validation
Paced	Yes	V3F: No direct term in recovered stage. V3AG: Direct subset and internal panels documented.	No native P6/P8 endpoint	Unchanged V3AG global output with new threshold; no new head. Held PTB-XL fold-9 contrast after folds 1-8 threshold fitting.	New operating point; incomplete historical ancestry; not a native specialist decision
LBBB	Yes	V3F: No direct term in recovered stage. V3AG: Not in the recovered eight-target subset.	No native P6/P8 endpoint	New SPH binary head evaluated. Complete-LBBB mapped contrast.	Complete and incomplete block are distinct references
2AVB	Yes	V3F: No direct term in recovered stage. V3AG: Not in the recovered eight-target subset.	No native P6/P8 endpoint	New SPH binary head; separate V3AG rank-only analysis. Source-specific binary fit; pooled PTB-XL ranks descriptive.	Held V3AG binary estimate not supported by one positive subject
3AVB	Yes	V3F: No direct term in recovered stage. V3AG: Not in the recovered eight-target subset.	No native P6/P8 endpoint	No supported SPH binary fit; V3AG rank-only analysis. Pooled PTB-XL ranks and positive-only LUDB references.	Held V3AG binary estimate not supported by two positive subjects
LQTS	Yes	V3F: No direct term in recovered stage. V3AG: Not in the recovered eight-target subset.	No native P6/P8 endpoint	Not evaluable with compatible direct syndrome references. No authorized compatible direct reference in this evaluation.	Prolonged QT is not a substitute for congenital LQTS
ALMI	Yes	V3F: Outside recovered P4 direct objective. V3AG: Outside recovered P4 direct objective.	No native P4 endpoint	Separate PTB-XL-trained experimental head evaluated. Frozen PTB Diagnostic transport; original auxiliary rank separate.	Not the published P4 head; limited transported sensitivity
ILMI	Yes	V3F: Outside recovered P4 direct objective. V3AG: Outside recovered P4 direct objective.	No native P4 endpoint	Separate PTB-XL-trained experimental head evaluated. Frozen PTB Diagnostic transport; original auxiliary rank separate.	Inferior and inferolateral localization remain distinct
LAE	Yes	V3F: Outside recovered P4 direct objective. V3AG: Outside recovered P4 direct objective.	No native P4 endpoint	Original auxiliary rank only; new LAO/LAE union head. Explicit SPH LAE rank contrast; separate PTB-XL union.	Union evidence does not validate isolated clinical LAE or equate LAO with LAE

Evidence boundary. Direct supervision means a documented objective in the recovered stage, not a fully authenticated training roster or complete pretraining history. Global-logit presence does not establish direct label exposure, a specialist decision or clinical validation. NSR is normal rhythm. The P6/P8 residual unions and the ALMI/ILMI/LAO–LAE pathology union are distinct experimental endpoints; their results do not validate every component individually. No published all-class Other specialist is established. Unrelated 41M experiments are not attributed to the released 58.35M backbones.

Sources. See the linked output contract, supervision audit, and endpoint evidence. The companion CSV retains repository-relative source paths.